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Reviews

Haruo Kubozono (ed.) (2017). *The phonetics and phonology of geminate consonants*. (Oxford Studies in Phonology and Phonetics 2.) Oxford: Oxford University Press. Pp. x + 408.

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The phonetics and phonology of geminate consonants is a collection of 14 papers which grew out of a series of conferences on Japanese and other languages held in the period 2011–15.* The book is divided into two sections, roughly corresponding to phonetics and phonology respectively, although many of the papers involve both. Part I contains seven experimental papers examining articulatory, acoustic and perceptual properties of geminates and geminacy contrasts. The data come from Japanese, Italian, Russian, Norwegian and Tashlhiyt Berber, as well as perceptual data from English speakers. One paper examines children's acquisition of length contrasts, and another college-age speakers' L2 acquisition. Part II contains seven papers that are more of a mixed bag methodologically: we find experiments using lexical priming, production data and orthographic judgements, qualitative descriptions of several languages, and a bit of optimality-theoretic analysis. The papers in this section are also more varied in terms of their subject matter: three papers address loanword adaptation and its relation to non-native perception, two are concerned with phonological processing and representations, one is a typological survey and one a detailed description and analysis of a particular language.

The book is excellent. One sometimes worries, with edited volumes, that people will send in 'B' material they can't publish elsewhere, or that the editors will solicit contributions from their friends, creating an ideological echo-chamber. This is not the case with the current volume: the papers here are of high quality, and most of them could be published in excellent linguistics journals. The papers cover a wide range of topics, and there is no dominant ideology here beyond the idea that details matter. Of course, particular authors do argue for particular representations and processes, but the spirit overall is one of pluralism with regard to both methodology and theoretical orientation. It is important reading for anyone who works on the phonology of geminates, but its appeal is far wider than that might suggest. Because of the thorny empirical and theoretical issues raised by length contrasts, in particular concerning the relationship between phonetic properties and abstract phonological ones, this book will be interesting to anyone concerned with the boundary areas between phonetics and phonology.

Rather than summarising every paper in the limited space here, I will follow some thematic threads that run through the collection, and try to draw some lessons from them. The following sections are concerned with the phonetic

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patterning of duration, the nature of symbolic representations and the division of labour between phonetically grounded and abstract analysis.

1 Everything interacts with everything

When I told my advisor that I wanted to write a dissertation about featural and prosodic influences on segment duration, he asked whether I was sure, and warned me that when it comes to duration, 'everything interacts with everything'. A partial list of factors that can influence consonant duration includes manner, place and laryngeal features, local segmental context, and local and global patterns of prosodic prominence and grouping. Given that all of these factors can affect duration, it follows that variation in duration could potentially be a cue to any of them in perception. In addition, some or all of these interactions will differ in their presence and magnitude across languages. Because length contrasts are primarily associated with duration, their production and perception should also interact with 'everything', and this is one theme running through the current volume: the production and perception of length contrasts are sensitive to a wide range of phonetic and phonological factors that differ between languages.

Olga Dmitrieva's contribution examines geminates in Russian, a language where length contrasts are not robustly maintained in casual speech. She shows that acoustic duration and neutralisation rates of singletons and geminates vary by segmental features, position in the word, adjacent segments and position with regard to stress. There is a somewhat surprising lack of correlation here between the probability of the neutralisation of a length contrast and the difference in acoustic duration present when that contrast is maintained. A number of the results here also seem to run contrary to typological tendencies noted by previous authors; for instance, voiced obstruents are realised as geminates more often than other segment types, despite their aerodynamic and cross-linguistic markedness (see Hayes & Steriade 2004 for an overview). Dmitrieva suggests that typological asymmetries are due to factors besides differences in relative duration, and that adjustments to relative duration occur on a language-specific basis for enhancement. And indeed, most hypotheses about such licensing asymmetries don't posit relative duration itself as the driving phonetic force, but rather differences in the *perceptibility* of duration. Shigeto Kawahara & Melanie Pangilinan, for instance, show that English speakers are much better at discriminating short–long pairs of non-speech sounds when those sounds mimic the gross spectral and intensity contours of stops than fricatives, and also better with fricative-like stimuli than approximant-like ones. This is despite the fact that duration differences were equal for all stimulus pairs, and despite English speakers' ostensible lack of experience with phonological length contrasts.

Hajime Takeyasu & Mikio Giriko's contribution focuses on within-category *vs.* between-category durational variation in Japanese and its effects on perception. Along the way, they help clear up a mystery concerning inconsistent results in the previous literature: some studies have found that lengthening a preceding vowel results in fewer long-consonant responses to an ambiguous stimulus (a CONTRASTIVE effect), while other studies report *more* long-consonant responses (an ASSIMILATIVE effect). The authors reproduce both effects here: the contrastive effect is triggered by large amounts of vowel lengthening, which results in the perception of phonologically long vowels, and activates a gradient phonotactic constraint in Japanese against sequences of long vowel followed by long consonant;

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the assimilative effect is triggered by smaller amounts of vowel lengthening, which are perceived as within-category phonetic variation and interpreted in accordance with Japanese phonetic variation, with longer vowels before long consonants than short ones. The study is a textbook illustration of how multiple factors may affect the perception of length, generating disparate and in some cases opposite effects.

Elinor Payne, Brechtje Post, Nina Gram Garmann & Hanne Gram Simonsen's paper adds age to the list of factors that affect duration and length contrasts. They examine the acquisition of length contrasts in Norwegian. Linguists have long debated whether Norwegian stressed syllables have a contrast for consonant length with compensatory vowel shortening, or a contrast for vowel length with compensatory consonant shortening. If a pattern is this hard for linguists to figure out, it may well present a challenge to infants as well. And the results here show that Norwegian children aged 2;6 to 6;0 produce fairly stable contrasts in terms of vowel-to-consonant ratio, but not in terms of absolute consonant or vowel durations. One interpretation of this (not endorsed by the authors) would be that children have learned vowel and consonant length covariation in the requisite way without inferring anything about whether the vowel or the consonant carries the length contrast. Equally interesting here is a regression around the age of six: children begin to speak at a more adult-like fast rate, which in turn compresses the temporal 'space' they have available for expressing the length contrast and renders the contrast temporarily less robust. This is a type of U-shaped learning curve I have never seen before. The authors also show that Norwegian adults use preaspiration and/or breathiness as a cue to voiceless geminates, and that children learn to do the same quite early. This is one of several results in the book that suggests a connection between consonant length and 'more consonantal' consonants; Rachid Ridouane & Pierre A. Hallé's contribution, for instance, shows that the Tashlhiyt Berber contrast between short and long voiceless stops in absolute initial position, which is extremely difficult to perceive, is implemented mainly through cues associated with the fortis or tense feature, including increased VOT and higher noise intensity for the long stops. Hyunsoon Kim's contribution shows that Korean speakers choose to represent Japanese geminates with symbols corresponding to tense/fortis stops, rather than lax/lenis or aspirated. It would be interesting to see a typological exploration of the association between consonantal length and lower sonority or increased consonantal strength.

2 Abstraction and moras

While most of the papers in Part I focus on ways in which phonetic properties may drive phonological patterning, several papers in Part II argue that some phonological properties are irreducibly abstract. Stuart Davis's analysis of Chuukese makes a strong case that MORAICITY is such a property. Although the facts are not framed this way in the paper, I find the following formulation to be useful. There is a class of phonological objects *C* in Chuukese that all bear some property *P*. Objects with property *P* are systematically: (i) deleted at the ends of words, and (ii) the only objects that count towards satisfying the word-minimality requirement. So far, *P* could be any normal feature, and we could stipulate that *P* happens to be what these patterns care about. But the class *C* includes (i) short vowels, (ii) whatever distinguishes long from short vowels and (iii) whatever distinguishes long from short consonants. Given the complete lack of unifying phonetic properties in *C*, it follows that *P* must be an abstract structural property,

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namely moraicity. While the presentation and conceptual analysis here are wonderful, some of the mechanics seem a bit suspect. The analysis entails a total lack of geminate-final URs in Chuukese, which violates the richness of the base assumption. It is fairly clear how such words ought to behave in different morphological contexts given Davis's analysis, so the fact that they apparently don't exist is troubling. That said, there is a lot to learn from this paper and whatever shortcomings it may have are more than made up for by its strengths.

Nina Topintzi & Stuart Davis argue that mora theory not only helps us organise heterogeneous collections of phonological objects that behave as natural classes within a language, but also makes it possible to see systematic implicational patterns across languages. Their typological survey of edge geminates, i.e. those found at the beginnings and endings of words, focuses on the relationship between the moraic status of geminates and the moraic status of consonant clusters. They propose the following implicational universal: if a language has edge geminates that are non-moraic, then it also has consonant clusters at that edge that are non-moraic; if a language has edge geminates that are moraic, this entails nothing about the availability or moraicity of consonant clusters at that edge. This is an interesting generalisation, although, as the authors note, it should be taken as provisional, given the relatively small number of languages described as having edge geminates. They propose that there are two different kinds of geminates cross-linguistically. Some are underlyingly moraic, yet occupy only one structural position; this implies nothing about the existence or moraicity of clusters, which have a different structural description. Other geminates are represented structurally like clusters and don't bear moras; a language that allows this configuration will also allow true clusters, which also won't bear moras.

The question of whether geminates differ from singletons in being underlyingly moraic also animates contributions by Sandra Kotzor, Allison Wetterlin & Aditi Lahiri and Lara Ehrenhofer, Adam C. Roberts, Sandra Kotzor, Allison Wetterlin & Aditi Lahiri. The first paper uses an ingenious semantic priming task to show that, for Bengali speakers, non-existing words with geminates activate their existing singleton counterparts, but non-existing words with singletons fail to activate their existing geminate counterparts. A slightly different way of putting this is that shortening a geminate interferes with lexical access in a way that lengthening a singleton does not. Activation is measured through decreased reaction times in a lexical decision task for a word semantically related to the existing counterpart of the presented stimulus, as well as increased negative-going magnitude of the N400 ERP signature associated with semantic expectedness. The second paper reports a very similar experiment in Swiss German. Here, non-existing words with short consonants *do* activate their existing counterparts with geminates. Arguably, the effect is not as large as the converse one from non-existing geminates to existing singletons, although the differences are small and the relevant interactions do not appear to have been directly assessed. Furthermore, the ERP results here appear to go in the opposite direction to the Bengali experiment and to expectations: in all conditions, the semantically related primes elicit more positive-going ERP signatures than the unrelated ones.

The authors interpret these experiments as showing that geminates are specified for something that singletons are not, which seems sensible. Their particular structural hypothesis, on the other hand, is less convincing. The claim is that geminates are underlyingly specified as moraic, rather than as occupying two consonantal timing slots. They suggest that the presence of a mora is privative, so increases in duration don't mismatch with the absence of a mora in a UR; while

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the presence of a second timing slot is not privative, so increases in duration *do* mismatch with the absence of a second structural position. I fail to see how this follows from the nature of the moraic or time-slot theories. That said, the results are clearly of interest and require some kind of an explanation. This leads me to the final thematic thread I will discuss here: the balance between 'concrete' phonetic and abstract phonological factors in explaining the perception, production and patterning of sounds.

3 Substance, abstraction and explanation

The two priming papers discussed above illustrate, among other things, just how tricky and flexible the mapping is between experimental data on the one hand and the computational-level description of sounds and their patterning on the other. While the authors clearly believe that two competing computational-level descriptions of geminate representation make different predictions about the processing data, it is not clear to me how these predictions actually follow from the representations in question. And as with most processing experiments, there may well be alternative explanations in terms of less abstract principles of production or perception. For instance, the finding that shortening a geminate interferes with lexical access, while lengthening a singleton does not, could be explained if singleton consonants display more durational variability in Bengali than geminate ones. If Bengali speakers have more experience with singletons being lengthened (through stress, prosodic-edge strengthening or just random variation) than with geminates being shortened, explaining the results would not require any recourse to abstract structural factors. And according to Kirchner's (1998) typological survey, one major class of shortening processes (prosodically conditioned lenition) never affects (surface) geminate consonants.

A second possibility is that the results here could be affected by the overall frequency of different types of singletons and geminates in Bengali. For instance, Shin'ichi Tanaka's paper shows that Japanese listeners are more likely to identify voiceless Italian obstruents as geminates than voiced ones, which are in turn more likely to be perceived as geminates than liquids, mirroring the Japanese-internal markedness scale. For liquids, Japanese subjects are more likely to mislabel geminates as singletons than *vice versa*. This converges nicely on the Bengali results: long consonants are more consistent with the singleton category than short ones are with the geminate category. But for voiceless obstruents, the effect is reversed: subjects are about twice as likely to label short consonants as geminates than *vice versa*. Neither of the priming papers breaks down results by consonant manner, nor do they discuss the frequency of different geminate and singleton types or their durational variability in Bengali or Swiss German. So it seems premature to draw any strong conclusions about abstract representations on the basis of these data. Tanaka's manner-dependent results also dovetail nicely with Kawahara & Pangilinan's contribution, which found similar asymmetries in English listeners' perception of duration contrasts for non-speech analogues of obstruents and sonorants.

The Tanaka paper also shows an interesting disconnect between Japanese listeners' perception of Italian length contrasts on the one hand and the computational-level properties of the loanword grammar on the other. Italian geminates are sometimes perceived as singletons, and sometimes appear as singletons when loaned into Japanese; however, Italian singletons are virtually never loaned into Japanese as

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geminates, despite the fact that Japanese listeners perceive them as geminates reasonably often. One lesson here is that, while phonetic factors may play a role in phonological generalisations, those generalisations must also be constrained by factors beyond 'raw' articulatory and perceptual data.

Junko Ito, Haruo Kubozono & Armin Mester's paper focuses on geminates in Japanese loanwords, giving a detailed and thorough OT analysis of prosodic constraints on their distribution. And thorny issues of abstract *vs*. phonetic explanation also arise in this paper. The authors argue for two somewhat unusual types of abstract structural constraint: prosodic markedness for feet, which can have a non-local influence on geminacy elsewhere in the word, and prosodic faithfulness to syllable position, a constraint type that is generally believed to be absent from native input–output phonology. The argument for foot markedness comes from mappings like English [lɪsən] *listen* to Japanese [risːun], as compared to English [lɪsənəɹ] *listener* to Japanese [risunaː]. The idea is that [s] lengthens in the first form because the final syllable is heavy, it therefore constitutes a foot by itself, the 'stranded' initial CV syllable is not heavy enough to be a foot and Japanese constrains words to begin with feet. In the second form, the addition of a third syllable means that a CVCV foot can be formed at the beginning of the word without geminating anything. This analysis seems plausible, but it's also an example of the difficulties that arise once we acknowledge the possibility of source-language phonetics influencing the adaptation of loanwords. Because 'everything interacts with everything', it is very hard to tell whether this type of mapping really requires hidden structure to be explained. For instance, English displays polysyllabic shortening, where syllables are shorter in words with more of them (Lehiste 1972). One could imagine that this shortening might affect Japanese listeners' perception of consonant length. Another possibly relevant fact discussed by Takeyasu & Giriko and by Yukari Hirata involves the relationship between consonant and word duration, which is a more reliable index of geminacy than local duration ratios. It is hard to say *a priori* how this property might be affected by English word-level duration adjustments, but it is at least a relevant cue to keep in mind. These phonetic explanations may not be particularly likely, but they serve to illustrate the general point that, with so many possible influences on duration and length perception, it is awfully hard to be confident in attributing particular grammatical patterns to abstract structural properties rather than phonetic ones. Along the same lines, Ito *et al*. suggest that prosodic faithfulness is necessary to explain adaptations such as English [maɹkət] *market* to Japanese [maːketːo]. The authors propose that the geminate realisation of [t] preserves its syllable-coda status from English. But Kubozono *et al*. (2013) show that English stops in this position display a range of durational and F0 properties that make Japanese listeners more likely to perceive them as geminates. So it's not clear that explaining this pattern requires the Japanese grammar to have access to hidden prosodic structure from the English source.

4 Summary

My attempt to draw connections between the first and second parts of this book is a response to the fact that, even though several of the contributions examine the interplay of phonetics and phonology in some detail, there is still something of a disconnect between the phonetic and phonological sections. This reflects the state of the field: one sometimes gets the disorienting feeling that, even with a

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large shared body of quantitative and qualitative data, phoneticians and phonologists can't agree on any of the computational-level properties of phonetic and phonological systems, nor even what the terms 'phonetics' and 'phonology' mean. In particular, the balance between phonetics and abstraction in explaining sound patterns is a core issue in the field with no general consensus. And this book certainly won't settle this debate, although some of the arguments for abstract analyses of particular phenomena are very strong. But by grappling with or at least implicating these difficult and ambiguous details of analysis, the book is elevated from a narrow examination of a specific topic to a work of much wider interest. The underlying theoretical issues at play here make it a worthwhile read for anybody concerned with the structure of, and relationship between, phonetics and phonology.

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